

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1

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TASK-2

1. TITLE

Implementation of HTML5 Canvas and CSS3 Styling

2. AIM

To create webpages using **HTML5 Canvas** for drawing graphical objects and **CSS3** for styling webpages using colors, gradients, navigation bars, cards, buttons, shadows and other styling properties.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the html5 <canvas> element.
- To draw shapes using javascript.
- To draw a rectangle, circle and line on a canvas.
- To display text on the canvas.
- To understand the canvas 2d rendering context.
- To apply css3 styling to an HTML webpage.
- To use CSS colors, gradients and fonts.
- To create a navigation bar using CSS.

- To create content cards using Flexbox.
 - To apply borders, border-radius and box shadows.
 - To use hover effects.
 - To style buttons and footer sections.
-

4. THEORY

A. HTML5 Canvas

The HTML5 <canvas> element is used to draw graphics dynamically on a webpage using JavaScript.

The canvas provides a drawing area with a specified width and height.

The following JavaScript code is used to access the canvas:

```
var canvas = document.getElementById("myCanvas");
```

```
var ctx = canvas.getContext("2d");
```

Here:

- canvas refers to the HTML canvas element.
- getContext("2d") provides a 2D drawing environment.
- ctx is used to draw shapes and text.

Canvas Functions Used

fillRect()

Draws a filled rectangle.

beginPath()

Starts a new drawing path.

arc()

Draws a circle or arc.

moveTo()

Sets the starting point of a line.

lineTo()

Creates a line from the current point to a specified point.

stroke()

Draws the outline of the current path.

fillText()

Displays text on the canvas.

B. CSS3 Styling

CSS stands for **Cascading Style Sheets**. It is used to control the appearance and layout of HTML elements.

The second program demonstrates several CSS3 properties including:

- Background colors
- Linear gradients
- Text shadows
- Font styling
- Navigation styling
- Hover effects
- Flexbox
- Borders
- Rounded corners
- Box shadows
- Button styling
- Footer styling

The display: flex property is used to arrange the cards horizontally.

The :hover pseudo-class is used to change the appearance of elements when the mouse pointer moves over them.

5. ALGORITHM / PROCEDURE

Program 1 – HTML5 Canvas Create an HTML5 document.

1. Add a heading to the webpage.
2. Create a canvas using the element.
3. Set the canvas width and height.
4. Access the canvas using JavaScript.
5. Create a 2D drawing context.
6. Draw a rectangle using fillRect().
7. Draw a circle using arc().
8. Draw a line using moveTo() and lineTo().
9. Display text using fillText().
10. Open the webpage in a browser.
11. Verify the graphical output.

Program 2 – CSS3 Styling Create an HTML5 document.

1. Add CSS inside the <style> element.
 2. Style the webpage body.
 3. Create a header with a gradient background.
 4. Create and style the navigation bar.
 5. Add hover effects to navigation links.
 6. Create three content cards.
 7. Use Flexbox to arrange the cards.
 8. Apply borders, rounded corners and shadows to the cards.
 9. Style the buttons.
 10. Add button hover effects.
 11. Create and style the footer.
 12. Open the webpage in a browser.
 13. Verify the styling and layout.
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6. PROGRAM

canvas.html

```
<!DOCTYPE html>
<html lang="en">

<head>

  <meta charset="UTF-8">

  <title>HTML5 Canvas</title>

</head>
```

```
<body>

<h1>HTML5 Canvas Example</h1>

<canvas
  id="myCanvas"
  width="500"
  height="300"
  style="border:1px solid black;"
>
  Your browser does not support
  the canvas element.
</canvas>

<script>

  var canvas = document.getElementById("myCanvas");

  var ctx = canvas.getContext("2d");


  // Rectangle

  ctx.fillStyle = "lightblue";

  ctx.fillRect(20, 20, 120, 70);


  // Circle

  ctx.beginPath();

  ctx.arc(
    220,
    55,
    35,
    0,
    2 * Math.PI
  );

  ctx.stroke();


  // Line

  ctx.beginPath();

  ctx.moveTo(20, 150);

  ctx.lineTo(250, 150);

  ctx.stroke();
```

```
// Text

ctx.font = "24px Arial";

ctx.fillStyle = "blue";

ctx.fillText(
    "Mohammed Almas",
    20,
    230
);

</script>

</body>

</html>
```

Css3.html

```
<!DOCTYPE html>
<html lang="en">

<head>

<meta charset="UTF-8">

<title>CSS3 Styling</title>

<style>

body {
    margin: 0;
    font-family: Arial, sans-serif;
    background: #f4f4f4;
}

/* Header */

header {
    background: linear-gradient(
        to right,
        #4facfe,
        #00f2fe
    );

    color: white;
    text-align: center;
    padding: 20px;
```

```
text-shadow: 2px 2px 4px gray;

font-size: 30px;
}

/* Navigation */

nav {
  background: #333;
  text-align: center;
  padding: 12px;
}

nav a {
  color: white;
  text-decoration: none;
  margin: 20px;
  font-size: 18px;
}

nav a:hover {
  color: yellow;
}

/* Cards */

.container {
  display: flex;
  justify-content: space-around;
  margin: 20px;
}

.card {
  width: 28%;
  background: white;

  border: 2px solid #ddd;
  border-radius: 10px;

  padding: 20px;

  box-shadow: 5px 5px 10px gray;

  text-align: center;
}

.card h2 {
  font-family: Georgia, serif;
}
```

```
.card button {  
    background: #007BFF;  
  
    color: white;  
  
    border: none;  
  
    padding: 10px 20px;  
  
    border-radius: 5px;  
  
    cursor: pointer;  
}
```

```
.card button:hover {  
    background: green;  
}
```

```
/* Footer */
```

```
footer {  
    background: #222;  
  
    color: white;  
  
    text-align: center;  
  
    padding: 15px;  
  
    margin-top: 20px;  
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<header>
```

```
    CSS3 Webpage Design
```

```
</header>
```

```
<nav>
```

```
    <a href="#">Home</a>
```

```
    <a href="#">About</a>
```



```
<a href="#">Services</a>
```

```
<a href="#">Contact</a>
```

```
</nav>
```

```
<div class="container">
```

```
<div class="card">
```

```
<h2>Card 1</h2>
```

```
<p>
```

```
  This is the first content card.
```

```
</p>
```

```
<button>
```

```
  Read More
```

```
</button>
```

```
</div>
```

```
<div class="card">
```

```
<h2>Card 2</h2>
```

```
<p>
```

```
  This is the second content card.
```

```
</p>
```

```
<button>
```

```
  Read More
```

```
</button>
```

```
</div>
```

```
<div class="card">
```

```
<h2>Card 3</h2>
```

```
<p>
```

```
  This is the third content card.
```

```
</p>
```

```
<button>
```

```
  Read More
```

```
</button>
```

```
</div>
```

```
</div>

<footer>
  © 2026 CSS3 Styling Example
</footer>

</body>
</html>
```

7. CODE EXPLANATION

Code	Explanation
Code	Purpose
<canvas>	Creates a drawing area.
id="myCanvas"	Gives an identifier to the canvas.
width="500"	Sets canvas width.
height="300"	Sets canvas height.
getElementById()	Accesses the canvas element.
getContext("2d")	Creates a 2D drawing context.
fillStyle	Sets the fill color.
fillRect()	Draws a filled rectangle.

<code>beginPath()</code>	Starts a new path.
<code>arc()</code>	Draws a circle.
<code>moveTo()</code>	Specifies the starting point of a line.
<code>lineTo()</code>	Specifies the ending point of a line.
<code>stroke()</code>	Draws the path outline.
<code>font</code>	Sets the font style and size.
<code>fillText()</code>	Displays text on the canvas.

B. CSS3 Program

CSS Property	Purpose
<code>margin</code>	Controls the outer spacing.
<code>font-family</code>	Specifies the font.
<code>background</code>	Sets the background.
<code>linear-gradient()</code>	Creates a gradient background.
<code>color</code>	Sets text color.
<code>text-align</code>	Aligns text.

8. TEST CASES AND EXECUTION DATA

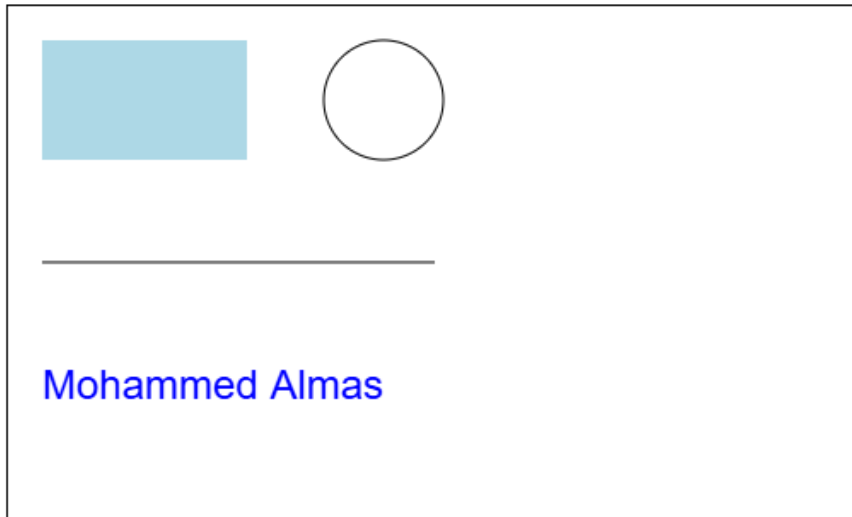
Test Case	Operation	Expected Result	Actual Result	Status
TC-01	Open canvas.html	Canvas should be displayed	Displayed successfully	Pass
TC-02	Draw rectangle	Light blue rectangle should appear	Displayed correctly	Pass
TC-03	Draw circle	Circle should appear	Displayed correctly	Pass
TC-04	Draw line	Horizontal line should appear	Displayed correctly	Pass
TC-05	Display text	Mohammed Almas should appear	Displayed correctly	Pass
TC-06	Open css3.html	Styled webpage should appear	Displayed successfully	Pass
TC-07	Check navigation	Navigation links should appear	Displayed correctly	Pass
TC-08	Check cards	Three cards should appear	Displayed correctly	Pass
TC-09	Hover navigation	Link color should change	Changed correctly	Pass
TC-10	Hover button	Button background should change	Changed correctly	Pass
TC-11	Check footer	Footer should appear at bottom	Displayed correctly	Pass

9. OUTPUT

A. canva.html

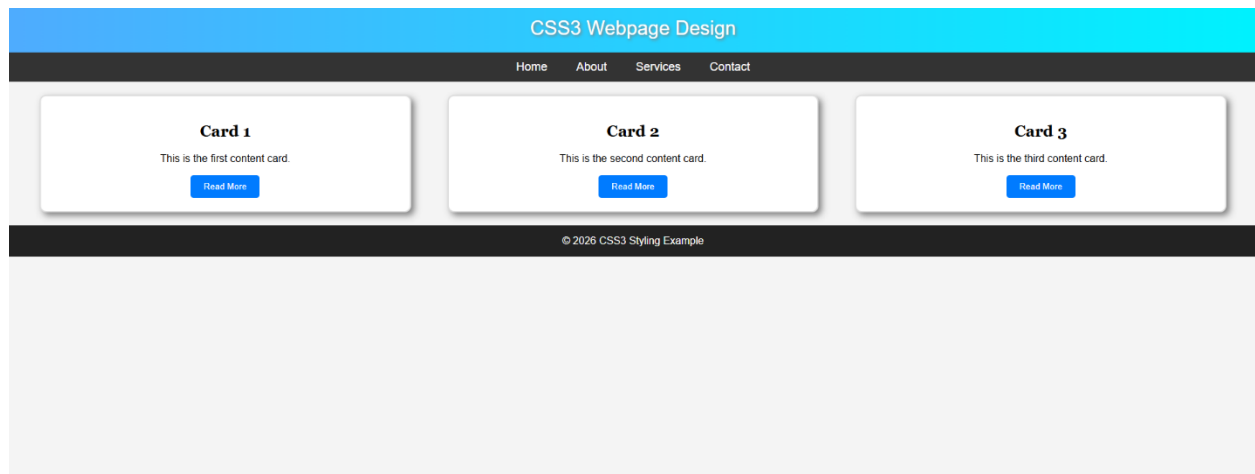
Output:

HTML5 Canvas Example



B. css3.html

Output:



10. RESULT

Thus, the **HTML5 Canvas and CSS3 styling programs** were successfully implemented and executed.

The Canvas program successfully demonstrated drawing a **rectangle, circle, line and text** using JavaScript. The CSS3 program successfully demonstrated **gradients, navigation styling, Flexbox, cards, buttons, shadows, rounded corners and hover effects**.

Hence, the required HTML5 Canvas and CSS3 concepts were successfully implemented and verified.
